

//SIT 55 : ASSIGNMENT 3 : INHERITANCE

/*Design and develop inheritance for a given case study, identify objects and relationships and implement inheritance wherever applicable.

Employee class has Emp_name, Emp_id, Address,Mail_id, and Mobile_no as members. Inherit the classes:

Programmer, Team Lead, Assistant Project Manager and Project Manager from employee class. Add Basic

Pay (BP) as the member of all the

inherited classes with 97% of BP as DA, 10 % of BP as HRA, 12% of BP as PF, 0.1% of BP for staff club

fund. Generate pay slips for the employees with their gross and net salary.

package Javaprogs;

class Employee

{

String Emp_name,Address,Mail_id;

long Emp_id;

long Mobile_no;

double DA,HRA,PF,SCFUND;

/*DA=Dearness Allowance ; HRA=House Rent Allowance ; PF=Provident Fund ;

SCFUND=Staff Club Fund*/

}

class Programmer extends Employee /*Inheritance achieved using "extends" keyword.

All attributes and methods , etc. of superclass can be accessed*/

{

double BPP,GROSS,NET;

Programmer(String name,String adr,String mid,long eid,long no,double BPP)

//Parameterized constructor;//this is reference variable which refers current object

{

this.Emp_name=name;

this.Address=adr;

this.Mail_id=mid;

this.Emp_id=eid;

this.Mobile_no=no;

this.BPP=BPP;

}

void display()

{

DA=((97*BPP)/100);

HRA=((10*BPP)/100);

PF=((12*BPP)/100);

SCFUND=((0.1*BPP)/100);

GROSS=(BPP+DA+HRA+PF+SCFUND);

NET=(BPP+DA+HRA)-(PF+SCFUND);

System.out.println("SALARY DETAILS OF EMPLOYEE: \n"+"NAME = "+Emp_name+"\n"+

"BP = "+BPP+"\n"+"DA = "+DA+"\n"+"HRA = "+HRA+"\n"+

"PF = "+PF+"\n"+"SCFUND = "+SCFUND);

System.out.println("GROSS SALARY= "+GROSS);

System.out.println("NET SALARY= "+NET+"\n");

```

}
}
class TeamLead extends Employee
{
double BPT,GROSS,NET;
TeamLead(String name,String adr,String mid,long eid,long no,double BPT)
//Parameterized constructor
{
this.Emp_name=name;
this.Address=adr;
this.Mail_id=mid;
this.Emp_id=eid;
this.Mobile_no=no;
this.BPT=BPT;
}
void show()
{
DA=((97*BPT)/100);
HRA=((10*BPT)/100);
PF=((12*BPT)/100);
SCFUND=((0.1*BPT)/100);
GROSS=(BPT+DA+HRA+PF+SCFUND);
NET=(BPT+DA+HRA)-(PF+SCFUND);
System.out.println("SALARY DETAILS OF TEAMLEAD : \n"+"NAME = "+Emp_name+"\n"+
"BP = "+BPT+"\n"+"DA = "+DA+"\n"+"HRA = "+HRA+"\n"+
"PF = "+PF+"\n"+"SCFUND = "+SCFUND);
System.out.println("GROSS SALARY= "+GROSS);
System.out.println("NET SALARY = "+NET+"\n");
}
}
class APM extends Employee //Assistant Project Manager
{
double BPA,GROSS,NET;
APM(String name,String adr,String mid,long eid,long no,double BPA)
//Parameterized constructor
{
this.Emp_name=name;
this.Address=adr;
this.Mail_id=mid;
this.Emp_id=eid;
this.Mobile_no=no;
this.BPA=BPA;
}
void cal()
{
DA=((97*BPA)/100);
HRA=((10*BPA)/100);
PF=((12*BPA)/100);
SCFUND=((0.1*BPA)/100);

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GROSS=(BPA+DA+HRA+PF+SCFUND);
NET=(BPA+DA+HRA)-(PF+SCFUND);
System.out.println("SALARY DETAILS OF ASSISTANT PROJECT MANAGER: \n"+"NAME =
"+Emp_name+"\n"+
"BP = "+BPA+"\n"+"DA = "+DA+"\n"+"HRA = "+HRA+"\n"+
"PF = "+PF+"\n"+"SCFUND = "+SCFUND);
System.out.println("GROSS SALARY="+GROSS);
System.out.println("NET SALARY = "+NET+"\n");
}
}
class ProjectManager extends Employee
{
double BPPM,GROSS,NET;
ProjectManager(String name,String adr,String mid,long eid,long no,double
BPPM) //Parameterized constructor
{
this.Emp_name=name;
this.Address=adr;
this.Mail_id=mid;
this.Emp_id=eid;
this.Mobile_no=no;
this.BPPM=BPPM;
}
void sal()
{
DA=((97*BPPM)/100);
HRA=((10*BPPM)/100);
PF=((12*BPPM)/100);
SCFUND=((0.1*BPPM)/100);
GROSS=(BPPM+DA+HRA+PF+SCFUND);
NET=(BPPM+DA+HRA)-(PF+SCFUND);
System.out.println("SALARY DETAILS OF PROJECT MANAGER: \n"+"NAME = "+Emp_name+"\n"+"BP =
"+BPPM+"\n"+"DA = "+DA+"\n"+"HRA = "+HRA+"\n"+
"PF = "+PF+"\n"+"SCFUND = "+SCFUND);
System.out.println("GROSS SALARY "+GROSS);
System.out.println("NET SALARY = "+NET+"\n");
}
}
public class Inheritance
{
public static void main(String args[])
{
Programmer P=new
Programmer("ANIRUDDHA","A.NAGAR","anir@gmail.com",55,8600679223L,66320.22f); //object
creation
TeamLead T=new TeamLead("OM","PUNE","omi123@gmail.com",21,9123557895L,23230.20f);
APM A=new APM("SHYAM","MUMBAI","SHYAM@gmail.com",230,8123557895L,50000.00f);
ProjectManager M=new

```

```
ProjectManager("RAM","NASHIK","ramoffice@gmail.com",1,8188557895L,23230.20f);
P.display(); //method call
T.show();
A.cal();
M.sal();
}
}
```

/*OUTPUT:

SALARY DETAILS OF EMPLOYEE:

NAME = ANIRUDDHA
BP = 66320.21875
DA = 64330.6121875
HRA = 6632.021875
PF = 7958.42625
SCFUND = 66.32021875000001
GROSS SALARY= 145307.59928125
NET SALARY= 129258.10634375

SALARY DETAILS OF TEAMLEAD :

NAME = OM
BP = 23230.19921875
DA = 22533.2932421875
HRA = 2323.019921875
PF = 2787.62390625
SCFUND = 23.23019921875
GROSS SALARY= 50897.36648828125
NET SALARY = 45275.65827734375

SALARY DETAILS OF ASSISTANT PROJECT MANAGER:

NAME =SHYAM
BP = 50000.0
DA = 48500.0
HRA = 5000.0
PF = 6000.0
SCFUND = 50.0
GROSS SALARY=109550.0
NET SALARY = 97450.0

SALARY DETAILS OF PROJECT MANAGER:

NAME = RAM
BP =23230.19921875
DA = 22533.2932421875
HRA = 2323.019921875
PF = 2787.62390625
SCFUND = 23.23019921875
GROSS SALARY 50897.36648828125
NET SALARY = 45275.65827734375*/